

$tg\alpha = \frac{1000}{1} =$

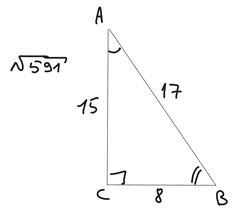
$= 1000$

$0 < \sin\alpha < 1$

$0 < \cos\alpha < 1$

$tg\alpha > 0$

№591



По т. Пифагора:

$17 \cdot 17 = 64 + AC^2$

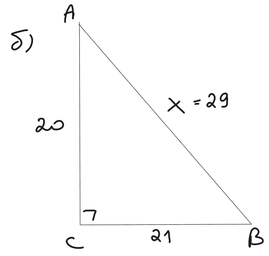
$289 = 64 + AC^2$

$AC^2 = 289 - 64$

$AC^2 = 225$

$AC = \sqrt{225} = 15$

б)



По т. Пифагора:

$x^2 = 20^2 + 21^2$

$x^2 = 841$

$x = 29$

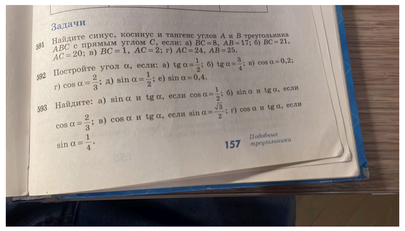
$\sin A$	$\cos A$	$tg A$
$\sin B$	$\cos B$	$tg B$

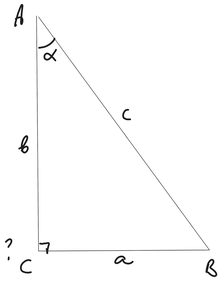
$\sin A = \frac{21}{29}$ $\quad \cos A = \frac{20}{29}$ $\quad tg A = \frac{21}{20}$

$\sin B = \frac{20}{29}$ $\quad \cos B = \frac{21}{29}$ $\quad tg B = \frac{20}{21}$

Геометрия

№591 б) г)





$\sin\alpha = \frac{\text{против. катет}}{\text{гипотенуза}} = \frac{CB}{AB}$

$\cos\alpha = \frac{\text{прилежн. катет}}{\text{гипотенуза}} = \frac{AC}{AB}$

$tg\alpha = \frac{\text{против. катет}}{\text{прилежн. катет}} = \frac{CB}{AC}$

$ctg\alpha = \frac{\text{прилежн. катет}}{\text{против. катет}} = (tg\alpha)^{-1} = \frac{1}{tg\alpha}$

$\sin A = ?$ $\quad \cos A = ?$ $\quad tg A = ?$

$\sin B = ?$ $\quad \cos B = ?$ $\quad tg B = ?$

$tg\alpha = \frac{\text{против. катет}}{\text{прилежн. катет}} \; /: \text{гипотенуза} = \frac{\frac{\text{против. катет}}{\text{гипотенуза}}}{\frac{\text{прилежн. катет}}{\text{гипотенуза}}} = \frac{\sin\alpha}{\cos\alpha}$

$\sin A = \frac{8}{17}$ $\quad \cos A = \frac{15}{17}$

$\sin B = \frac{15}{17}$ $\quad \cos B = \frac{8}{17}$

$tg A = \frac{8}{15}$ $\quad \angle A + \angle B = 90°$

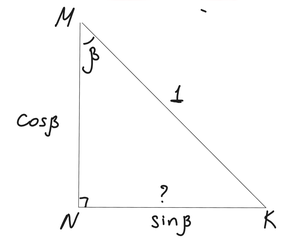
$tg B = \frac{15}{8}$ $\quad \angle A = 90° - \angle B$

$\sin\angle A = \cos\angle B$

$\sin(90° - \angle B) = \cos\angle B$

$\cos(90° - \angle B) = \sin\angle B$

$tg\alpha = \frac{\sin\alpha}{\cos\alpha}$



$\sin\beta = \frac{NK}{MK}$; $\sin\beta = \frac{NK}{1}$; $NK = \sin\beta$

$\cos\beta = \frac{MN}{MK}$; $\cos\beta = \frac{MN}{1}$; $MN = \cos\beta$

По теореме Пифагора:



$MK^2 = MN^2 + NK^2$

$1^2 = (\cos\beta)^2 + (\sin\beta)^2$

$(\sin\beta)^2 + (\cos\beta)^2 = 1$ – основное тригонометрическое тождество

$0 < \beta < 90°$

$\sin^2\beta = (\sin\beta)^2$

$\sin^2\beta^2 = (\sin\beta^2)^2$

$\sin^2\beta = (\sin\beta)^2$

$\sin^2\beta + \cos^2\beta = 1$

$\sin\alpha = \frac{1}{2}$

$\cos\alpha = ?$

Воспольз. ОТТ:

$\sin^2\alpha + \cos^2\alpha = 1$

$\left(\frac{1}{2}\right)^2 + \cos^2\alpha = 1$

$\cos^2\alpha = 1 - \left(\frac{1}{2}\right)^2$

$\cos^2\alpha = \frac{1}{1}^{(4} - \frac{1}{4}^{(1}$

$\cos^2\alpha = \frac{4}{4} - \frac{1}{4} = \frac{3}{4}$

$\cos^2\alpha = \frac{3}{4}$

$\cos\alpha = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2} \approx \frac{1{,}7}{2} \approx 0{,}85 < 1$

$\cos\alpha = -\sqrt{\frac{3}{4}} = -\frac{\sqrt{3}}{2}$ – отр. число не может быть отн. катета к гипот.